

VICHRUTH M

ML Engineer · Computer Vision Research · Edge AI · AI Automation

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Tiruppatur, TN, India

SUMMARY

- CSE undergraduate (VIT Vellore, 2024–2028); research internships at two national institutes, a first-author paper pending publication in an IEEE journal, and a live 24/7 production system run for a freelance client. Specialty: making heavyweight models run reliably on constrained edge hardware.

RESEARCH EXPERIENCE

Research Intern — LLM & Multimodal AI, NIT Tiruchirappalli (Supervisor: Dr. B. Janet) Jun – Jul 2026

- Built a multimodal classroom engagement-detection system running **fully offline on a Raspberry Pi 8 GB**: interval-based CV pipeline (ONNX face detection, engagement classifier) fused with teacher audio transcription, aggregated into a structured matrix fed to a 4-bit quantised local LLM.
- Benchmarked the edge system against a cloud VLM baseline to quantify the accuracy–latency trade-off; evaluated face-detection backbones (MediaPipe, YOLOv8n-face, UltraFace) for 70-student overhead-camera classrooms. Findings feeding a journal paper in preparation.

Research Intern — Computer Vision & Model Compression, IIIT Kottayam (Supervisor: Dr. Sridhar Raj S) May – Jun 2026

- Built **ViT-MTL**: TransReID backbone with a five-branch multi-task head (identity + soft-biometric attributes) and GRL domain adversarial training, trained end-to-end **within a 6 GB VRAM budget** (FP16 mixed precision, gradient accumulation, PyTorch 2.5).
- Diagnosed three silent failures (tensor-shape extraction, evaluation feature mismatch, missing identity supervision) that suppressed Rank-1 to 0.3% without crashing; fixes put accuracy on a convergence trajectory toward 55–68% on Market-1501.
- First-author paper pending publication in an IEEE journal.

INDUSTRY & FREELANCE EXPERIENCE

Freelance AI & Automation Engineer — Independent Client Projects (Remote) 2025 – Present

- Operate a **live 24/7 B2B lead-generation pipeline**: 8-phase automation (scrape → AI website audit via GPT-4o vision → personalised outreach → reply classification → proposals → Stripe payment → onboarding) on self-hosted n8n + PostgreSQL with 15+ API integrations, managed end-to-end from a phone via Telegram approval flows.

Machine Learning Intern — KalkiNi AI Surveillance Systems, Vellore Feb – Sep 2025

- Built multi-threaded video-stream pipelines (Pandas/NumPy) cutting preprocessing time ~35%; trained deep-learning classifiers for real-time threat profiling on live CCTV feeds using shared-feature multi-task transfer learning.

AI Intern — Cloudstier Solutions Pvt. Ltd., Tiruppatur May – Jun 2025

- Deployed an on-premise NLP chatbot (Django + Microsoft Phi SLM); handled supervised fine-tuning, prompt alignment, structured output, and REST endpoint validation.

SELECTED PROJECTS

NeuroLog — Offline Semantic Video Search PyTorch · CLIP · FAISS · OpenCV 2025

- Zero-shot multimodal video search fully offline in 6 GB VRAM; ~0.284 s query latency over a 1,414-frame FAISS index. Manually cast weights/tensors to FP16, cutting GPU memory ~1.5 GB → ~600 MB with retrieval accuracy intact.

NIKI-LAUDA-AI — Real-Time Race Engineering Pipeline Python · UDP · PyTorch · WINE Mar 2026 –

- Decodes 50 Hz binary C-struct telemetry over UDP from a physics simulator (run natively on Linux via WINE); training an LSTM on sliding-window time series for predictive tyre-degradation modelling.

Low-Parameter AI Bug Detector CodeT5 · LoRA/PEFT · Flask Jul – Oct 2025

- Fine-tuned CodeT5-base on a 2,000-sample bug-fix dataset with parameter-efficient LoRA for fully offline inference, served via a Flask web UI.

PUBLICATIONS & COMPETITIONS

- “Person Re-Identification using Multi-Task Vision Transformers with Domain Adversarial Training”** — Vichruth M, Dr. Sridhar Raj S (IIIT Kottayam, 2026). Pending publication, IEEE journal.
- 2nd Prize**, Hackademia, graVITas '25 — FFCS-Buddy, NLP course search (MiniLM + FastAPI) · **2nd Runner-Up**, National Ideathon 2024 — “Pre-Cog Brakes” predictive braking concept.
- Kaggle March ML Mania 2026 (leakage-safe expanding-window features) · FREUID Challenge 2026 (IJCAI-ECAI), ID-document fraud detection — active.

TECHNICAL SKILLS

- Languages**: Python (advanced), C++ (intermediate), Java, JavaScript, SQL
- ML / CV**: PyTorch, TensorFlow, Hugging Face, scikit-learn, XGBoost, CLIP, TransReID, LoRA/PEFT, OpenCV, FAISS, ONNX Runtime, YOLOv8, MediaPipe
- Edge AI**: FP16/INT8/4-bit quantization, pruning, llama.cpp, TinyML, Raspberry Pi, mixed-precision + gradient-accumulation training under tight VRAM
- Automation / Backend**: n8n (self-hosted), FastAPI, Flask, Django, Node.js, PostgreSQL, REST, Stripe, Telegram bots, Git, Linux, Docker

EDUCATION & LEADERSHIP

B.Tech CSE — Vellore Institute of Technology (VIT), Vellore · CGPA: 7.54/10 Jul 2024 – Jul 2028

- Senior Secondary (CBSE) — Maharishi Intl. Residential School, Chennai, 85% (2024) · Junior Core, The Electronics Club (TinyML, embedded C++ mentoring) · App Developer, Padasalai · Campus Ambassador, Masters' Union.